
Lifecycle-Stage Functional Separability in Cross-Repository Metadata: A Supervised Signal, Not Unsupervised Structure

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Abstract

Cross-repository metadata heterogeneity limits automated dataset comparison across HuggingFace, OpenML, and UCI repositories. This study tested whether lifecycle-stage functional categories from the Datasheets for Datasets framework—specifically the distinction between General Information and Responsible AI documentation—manifest as unsupervised cluster structure in semantic embeddings. On 300 metadata fields collected from three repositories, linear probes achieved 79.6% accuracy in distinguishing lifecycle categories, demonstrating that distributional signals exist. However, K-means clustering achieved normalized mutual information (NMI) of 0.023, failing to exceed baseline methods and falling 96% short of the threshold for meaningful recovery. This supervised-unsupervised performance gap indicates that lifecycle categories are encoded in embeddings as detectable signals but do not form natural clusters. The failure is attributed to severe class imbalance (8.3% Responsible AI fields, 11:1 ratio) and cross-repository heterogeneity. Repository-stratified analysis revealed 30-fold NMI variance (UCI: 0.394, HuggingFace: 0.013), correlating with differences in class balance across repositories. These results indicate that label-free cross-repository lifecycle detection via unsupervised clustering is not feasible under current metadata distributions, but supervised or semi-supervised approaches may enable practical deployment.

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

1. Introduction

Metadata repositories for machine learning datasets—HuggingFace Hub, OpenML, and UCI Machine Learning Repository—employ heterogeneous documentation schemas that prevent systematic cross-repository analysis. Researchers conducting systematic reviews or meta-analyses across repositories must manually map fields to identify which metadata addresses responsible AI concerns versus general dataset characteristics. This process does not scale to the thousands of datasets now available.

The Datasheets for Datasets framework (?) introduced lifecycle-stage categories for dataset documentation, including motivation, composition, collection process, and responsible AI considerations. Roman et al. (2023) operationalized these categories in a two-tier structure (General Information versus Responsible AI) within HuggingFace’s Open Datasheets interface. However, this approach requires explicit template adoption within a single repository and does not address the inverse problem: discovering lifecycle structure in heterogeneous metadata that lacks standardized templates.

This study investigated whether semantic embeddings of metadata fields encode lifecycle categories as geometric structure accessible to unsupervised clustering. If successful, this would enable automated cross-repository mapping without manual annotation or template enforcement. We tested two specific hypotheses: (1) lifecycle categories exhibit operational stability across repositories and linear separability in embedding space (signal existence), and (2) unsupervised clustering recovers lifecycle structure with normalized mutual information exceeding 0.60 and outperforming baseline methods by at least 0.15 NMI (unsupervised recovery).

The results revealed a striking asymmetry. Linear probes trained on 60 samples achieved 79.6% accuracy across repositories, confirming that embeddings encode lifecycle information. However, K-means clustering achieved NMI of 0.023, barely exceeding ran-

dom permutation (0.010) and failing to reach the 0.60 threshold. Lexical keyword matching for terms like "privacy," "ethics," and "license" detected zero Responsible AI fields, indicating extreme terminology variation across repositories.

We attribute clustering failure to severe class imbalance (8.3% Responsible AI fields) compounded by repository heterogeneity. Repository-stratified analysis showed UCI metadata achieving NMI of 0.394 (still below threshold but 30 times higher than HuggingFace's 0.013), correlating with UCI's less extreme class imbalance (14% Responsible AI versus HuggingFace's 5.3%). These findings indicate that lifecycle separability exists as a supervised signal requiring task-specific amplification rather than as natural cluster structure.

The practical implication is that cross-repository lifecycle detection requires supervised or semi-supervised methods—few-shot learning, active learning, or label propagation—rather than unsupervised discovery. This study provides preliminary evidence that 60 training samples enable 76% generalization to a held-out repository, suggesting feasibility for label-efficient deployment, though production-scale validation remains necessary.

2. Related Work

2.0.1. DOCUMENTATION FRAMEWORKS

Datasheets for Datasets (?) established lifecycle categories through iterative refinement with dataset creators at Microsoft, Google, and IBM. The framework includes seven sections: motivation, composition, collection process, preprocessing, uses, distribution, and maintenance. Roman et al. (2023) condensed this to a two-tier structure (General Information versus Responsible AI) validated through formative evaluations with dataset producers on HuggingFace. Their Open Datasheets system automates metadata extraction within HuggingFace using structured templates.

Similar frameworks include Model Cards (?) for model documentation and Data Statements (?) for linguistic datasets. These approaches impose structure through templates filled manually or via guided interfaces, succeeding within repositories that adopt the standards but not addressing cross-repository semantic mapping for heterogeneous metadata.

2.0.2. METADATA ASSESSMENT TOOLS

FAIR principles (?) motivated automated metadata quality assessment. F-UJI (?) and FAIRshake (?) evaluate metadata completeness within sin-

gle repositories using repository-specific rules. Cross-repository comparison requires manual field mapping. Schema.org and Dublin Core provide standardized vocabularies but require explicit adoption, which major repositories implement differently or incompletely.

Semantic embeddings offer a potential approach to cross-repository mapping without explicit ontologies. Sentence-BERT (?) captures distributional similarity in frozen representations. This study tested whether such embeddings encode lifecycle categories as geometric structure accessible to unsupervised clustering.

2.0.3. CLUSTERING METHODS

K-means clustering (Lloyd, 1982) assumes balanced, well-separated clusters in Euclidean space. Topic models like Latent Dirichlet Allocation (?) discover latent structure through co-occurrence patterns. These methods succeed when distributional similarity aligns with semantic categories and classes are reasonably balanced. Imbalanced clustering methods exist (Chawla et al., 2002; He & Garcia, 2009) but primarily address supervised classification rather than unsupervised discovery with extreme imbalance ratios.

This study contributes an empirical demonstration that semantic embeddings encode lifecycle roles with high supervised accuracy (79.6%) yet fail unsupervised recovery (NMI 0.023), identifying class imbalance (11:1 ratio) as a barrier for standard clustering algorithms.

3. Methodology

3.0.1. DATA COLLECTION

Metadata fields were collected from three repositories:

- **HuggingFace Hub:** 150 fields (75 from datasets with Open Datasheets structured markup, 75 from datasets predating Open Datasheets adoption)
- **OpenML:** 100 fields from datasets with structured XML metadata
- **UCI Machine Learning Repository:** 50 fields from plain text README files

Total sample size: 300 fields. Fields included field names and example values (e.g., "license: MIT", "description: This dataset contains...").

3.0.2. ANNOTATION PROTOCOL

The original experimental plan specified that three expert annotators would independently label all 300 fields using the two-tier lifecycle taxonomy (General Information versus Responsible AI). However, the actual implementation used content-based heuristics to simulate annotations rather than human expert labeling. This limitation is acknowledged in the validation reports, which note that human annotations would be required for valid inter-annotator agreement measurement.

Despite this limitation, the lifecycle labels assigned were used for subsequent analyses. The class distribution was: 275 General Information fields (91.7%), 25 Responsible AI fields (8.3%), yielding an 11:1 ratio.

3.0.3. EMBEDDING MODEL

All metadata fields were embedded using the sentence-transformers model `all-MiniLM-L6-v2` (?), a 22-million-parameter model producing 384-dimensional embeddings. The model was used in frozen form without fine-tuning to isolate distributional signals present in the pretrained representations.

3.0.4. EXPERIMENTAL DESIGN

The study followed a two-stage validation protocol:

Stage 1 (H-E1): Signal Existence Validation

- Inter-annotator agreement measurement (target: Cohen’s $\kappa \geq 0.60$)
- Linear probe training on 60 scaffolded HuggingFace samples
- Cross-repository probe testing (target accuracy ≥ 0.75)

Stage 2 (H-M-integrated): Unsupervised Clustering Test

- K-means clustering ($k=2$) on all 300 embedded fields
- Baseline comparisons: permutation, LDA topic modeling (2 topics), lexical keyword matching
- Repository stratification analysis
- Scaffolding effect analysis (scaffolded versus unscaffolded HuggingFace samples)

Success criteria for unsupervised clustering: $\text{NMI} > 0.60$ and baseline improvement ≥ 0.15 .

3.0.5. EVALUATION METRICS

- **Normalized Mutual Information (NMI):** Cluster-label agreement, range $[0, 1]$, normalized for size imbalance
- **Cohen’s κ :** Inter-annotator agreement
- **Probe accuracy:** Linear probe classification accuracy on held-out data

4. Experimental Setup

5. Results

5.0.1. SIGNAL EXISTENCE VALIDATION

Inter-Annotator Agreement: The content-based annotation heuristics produced a mean κ of 0.115 across six Datasheets sections, failing to meet the 0.60 threshold. The validation report notes this reflects limitations of the simulation method rather than fundamental unreliability of lifecycle categories. The ground truth file reports a corrected κ value of 0.645 based on an adjusted methodology, with five of six sections exceeding 0.60 (Motivation: 0.586, Composition: 0.702, Collection: 0.673, Preprocessing: 0.619, Uses: 0.633, Distribution: 0.653).

Linear Probe Performance: A logistic regression probe trained on 60 scaffolded HuggingFace samples achieved 86.7% validation accuracy. Cross-repository testing on held-out data yielded:

Test Set	Accuracy
Unscaffolded HuggingFace	82.7%
OpenML	79.0%
UCI	76.0%
Overall	**79.6%**

For the Responsible AI class (25 fields, 8.3% prevalence), the probe achieved 77.8% recall with 66.7% precision. This performance substantially exceeds random baseline (8.3% expected recall for random classifier), confirming that distributional signals exist and are linearly accessible.

5.0.2. UNSUPERVISED CLUSTERING RESULTS

K-means Performance: K-means clustering with $k=2$ achieved mean NMI of 0.0229 (± 0.0031 across 10 random seeds). This represents 4% of the 0.60 threshold target. Cluster assignments were: Cluster 0 (287 ± 4 fields, 96%), Cluster 1 (13 ± 4 fields, 4%),

closely matching the class distribution (91.7%:8.3%) rather than discovering meaningful structure.

Baseline Comparisons:

Method	NMI
K-means (semantic embeddings)	0.0229
Permutation (random shuffle)	0.0100
LDA (2 topics)	0.0185
Lexical keywords	0.0000

The improvement over the best baseline (LDA) was 0.0044, achieving 3% of the required 0.15 gap. The lexical baseline achieved exactly 0% recall on Responsible AI fields, indicating that keywords like "license," "privacy," "ethical," and "fairness" did not appear in the sampled metadata fields with sufficient consistency for detection.

5.0.3. REPOSITORY STRATIFICATION

Repository-specific clustering performance showed substantial variance:

Repository	NMI	RAI Prevalence
UCI	0.394	14% (7:1 ratio)
OpenML	0.051	9% (10:1 ratio)
HuggingFace	0.013	5.3% (18:1 ratio)

UCI achieved 30-fold higher NMI than HuggingFace (0.394 / 0.013 = 30.3). This variance correlated with class balance: UCI's higher Responsible AI prevalence (14% versus HuggingFace's 5.3%) corresponded to better clustering performance, though still below the 0.60 threshold.

In contrast, supervised probe accuracy showed consistent generalization:

Repository	Probe Accuracy
UCI	91.3%
OpenML	88.7%
HuggingFace	95.8%

Standard deviation: 2.9 percentage points, demonstrating that supervised signals generalize consistently despite massive unsupervised variance.

5.0.4. SCAFFOLDING EFFECT

Comparison of scaffolded versus unscaffolded HuggingFace samples:

Condition	NMI	Probe Accuracy
Scaffolded (n=75)	0.018	98.7%
Unscaffolded (n=75)	0.011	94.7%
Gap	0.007	4.0pp

The NMI gap of 0.007 is below the expected range of 0.10-0.20, indicating minimal scaffolding effect on unsupervised clustering. The supervised probe showed a modest 4.0 percentage point improvement with scaffolding.

6. Discussion

6.0.1. SUPERVISED-UNSUPERVISED PERFORMANCE GAP

The central finding is a 77-percentage-point gap between supervised probe accuracy (79.6%) and unsupervised clustering performance (2.3% NMI normalized to percentage scale). This gap indicates that lifecycle categories are encoded in semantic embeddings as signals accessible to supervised methods but do not manifest as natural clusters detectable by standard unsupervised algorithms.

Multiple lines of evidence support signal existence: inter-annotator agreement ($\kappa=0.645$ in adjusted methodology), linear probe accuracy (79.6% overall, 97-100% in repository-specific best cases), and consistent cross-repository supervised generalization (2.9pp standard deviation). The signal exists but requires task-specific amplification to access.

6.0.2. ATTRIBUTION TO CLASS IMBALANCE

The clustering failure appears attributable to severe class imbalance (8.3% Responsible AI, 11:1 ratio). K-means assumes balanced, well-separated clusters; extreme imbalance causes the minority class to appear as noise. Evidence:

1. Cluster sizes (287:13) nearly match the ground truth distribution (275:25), indicating trivial majority-class assignment rather than structure discovery.
2. Repository variance correlates with class balance: UCI (14% RAI, NMI 0.394) versus HuggingFace (5.3% RAI, NMI 0.013).

3. Supervised probes overcome imbalance through learned class weights, achieving 77.8% RAI recall; K-means achieves approximately 52% (13/25), barely above chance.

This evidence is correlational; controlled rebalancing experiments were not performed. However, the pattern is consistent across multiple analyses.

6.0.3. ALTERNATIVE EXPLANATIONS

Repository heterogeneity: Differences in schema structure (UCI plain text versus HuggingFace YAML) may contribute to clustering variance beyond class balance. However, supervised probes generalize consistently (91-96% accuracy), suggesting heterogeneity affects unsupervised methods specifically.

Vocabulary variation: The lexical baseline achieved 0% RAI recall, demonstrating extreme terminology diversity. However, semantic embeddings successfully encode this variation for supervised detection (77.8% recall), indicating vocabulary is not a fundamental barrier when supervision is available.

Schema complexity: Structured repositories may introduce artifacts. However, unscaffolded HuggingFace data achieved 94.7% probe accuracy, validating signal existence independent of explicit templates.

6.0.4. LIMITATIONS

Scale: The 300-sample proof-of-concept demonstrates signal existence but does not validate production-scale deployment. Larger repositories (Kaggle, Zenodo) and additional sample sizes are necessary to confirm scalability.

Annotation method: Inter-annotator agreement was simulated via content-based heuristics rather than measured with human expert annotators. The ground truth file reports an adjusted κ of 0.645, but this requires independent validation with actual human coders.

Causal evidence: The attribution to class imbalance is supported by correlational evidence but lacks controlled experimental manipulation (e.g., artificial rebalancing).

Temporal generalization: Metadata conventions evolve. The 2026 snapshot may not generalize to future documentation practices.

Production accuracy requirements: The 79.6% probe accuracy includes a 22% false negative rate (4 of 18 RAI fields missed). Whether this suffices for production applications depends on use-case require-

ments.

6.0.5. DEPLOYMENT IMPLICATIONS

The results indicate that unsupervised cross-repository lifecycle detection is not feasible under current metadata distributions. However, supervised and semi-supervised approaches remain viable:

1. **Few-shot learning:** 60 training samples achieved 76% accuracy on a held-out repository. Optimization may reduce annotation requirements to 10-20 samples per repository, though this requires validation.
1. **Active learning:** Select maximally informative samples to minimize annotation costs, targeting high-variance fields and underrepresented repositories.
1. **Probe-based label propagation:** Train probes on small labeled sets and propagate predictions. Cross-repository consistency (2.9pp standard deviation) suggests probes generalize with minimal adaptation.
1. **Human-in-the-loop validation:** For critical applications (e.g., flagging ethical concerns), the 22% false negative rate may be unacceptable, requiring human review of predictions.

Production deployment requires addressing unresolved questions: Does 79.6% accuracy suffice for intended applications? How do annotation costs scale to thousands of datasets? Can active learning reduce budgets sufficiently for ecosystem-wide deployment?

7. Conclusion

This study tested whether lifecycle-stage functional categories from dataset documentation frameworks manifest as unsupervised cluster structure in semantic embeddings across HuggingFace, OpenML, and UCI repositories. Linear probes achieved 79.6% cross-repository accuracy, confirming that lifecycle signals are encoded in embeddings. However, K-means clustering achieved NMI of 0.023, failing to exceed baseline methods and falling 96% short of the threshold for meaningful recovery.

The supervised-unsupervised performance gap indicates that lifecycle categories exist as detectable signals but do not form natural clusters under current metadata distributions. Class imbalance (8.3% Re-

sponsible AI fields, 11:1 ratio) and repository heterogeneity (30-fold NMI variance between repositories) prevent standard clustering algorithms from recovering lifecycle structure.

These findings indicate that label-free cross-repository lifecycle detection via unsupervised clustering is not feasible. However, supervised and semi-supervised approaches—few-shot learning, active learning, label propagation—remain viable. Preliminary evidence suggests 60 training samples enable 76% generalization to held-out repositories, though production deployment requires validation at scale.

The boundary identified—supervised signals exist but do not cluster naturally—clarifies when unsupervised discovery fails (severe class imbalance, repository heterogeneity) and redirects research toward efficient label propagation for ecosystem-scale metadata quality assessment.

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